INTERIM FEDERAL SPECIFICATION

FOUNTAIN PEN, TECHNICAL

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D.C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers tubular stylus type technical fountain pens, pen sets, and replacement point assemblies. The pens, sets, and replacement points are available in various line widths.

1.2 Classification.

1.2.1 Types and styles. Technical pens, including sets, shall be of the following types and styles, as specified (see 6.2).

Type I. Individual pen

Style 1: Pocket fountain type with built-in self-filling ink reservoir.
Style 2: Pocket fountain type with refillable ink reservoir (cartridge).

Type II. Sets (see 3.3.2).

1.2.2 Replacement point assemblies. (See 3.3.3 and 6.2.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-519 - Plastic Sheet; Tracing, Glazed and Matte Finish.
TT-I-528 - Ink, Drawing, Waterproof, Black.
TT-I-533 - Ink, Drawing, Waterproof, Black (For Use on Water Repellent and Plastic Drafting Surfaces).
UU-P-223 - Paper, Drawing.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Agent, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee, Agent:

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Bid samples. Bid samples, when required by the invitation for bid, shall be submitted in the quantities specified (see 6.2), and shall be examined in accordance with 6.4.

3.2 Materials. The manufacturer is given wide latitude in his selection of materials and process of manufacture provided the product meets the requirements of this specification.

3.3 Design and construction.

3.3.1 Type I (individual pen). Type I pens shall be furnished in the styles described in 1.2.1, and as specified (see 6.2). The style 1 pen shall be designed with a self-filling reservoir as an integral part of the barrel and point housing section; and shall consist of a barrel and point housing section, a cap with pocket clip, and a point section (see 3.3.3). The style 2 pen shall be designed with a refillable ink reservoir (cartridge) which shall attach to the point housing section by friction fit or with a locking clamp ring, and shall be enclosed within the barrel when the pen is assembled. The style 2 pen shall consist of a barrel, a cap with pocket clip, a reservoir (cartridge), a point housing section, and a point section (see 3.3.3). The pens shall be for use with drawing inks conforming to TT-I-528 and TT-I-533. The pens shall meet the features of 3.3.1.1, and the performance characteristics of 3.3.1.2. The pens shall be in sizes shown in table I, and as specified (see 6.2). Other commercial sizes may be specified provided the outside diameter of the point tip and tolerances are given (see 6.2). The writing tube or point shall be heat treated, chromium plated and polished stainless steel for general use; or tungsten carbide or synthetic jewel for use on plastic surfaces which have a matte finish, as specified (see 3.3.3 and 6.2). The cleansing wire for pens with tungsten carbide or synthetic jewel points shall be hard treated.

TABLE I

Size Designation	Outside Diameter At Point Tip
00	0.011 ± 0.001 inch
0	0.015 ± 0.002 inch
1	0.019 ± 0.002 inch
2	0.022 ± 0.002 inch
2½	0.028 ± 0.002 inch
3	0.037 ± 0.002 inch
3½ or 4	0.050 ± 0.002 inch

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3.3.1.1 Features. The pens shall have the following features:

The pens shall have a cap with a built-in liner molded to enclose the point tube to restrict air space around the pen point when the cap is in place.

The pens shall be provided with a color code so that the user may identify the different sizes. As a minimum, the color coding shall appear on the cap and point section.

The pens shall have an offset on the end of the point tube. (See 6.3.)

The pens shall have the line size designation (table I) conspicuously stamped on the pen barrel or cap. Additional millimeter marking is acceptable.

A point section wrench shall be provided with each pen.

3.3.1.2 Performance characteristics. The pens shall produce an unbroken, even line, without evidence of ink leakage around the point housing, point section, or cartridge fitting, when tested as specified in 4.4.1. After setting with cap in place and point up for 48 hours, the pens shall start producing an unbroken and even line within 15 seconds when tested as specified in 4.4.1.1.

3.3.2 Type II (sets). Sets shall consist of the style 2 pen described in 1.2.1 and 3.3.1, or may at the option of the supplier, consist of drawing point section pens described in 3.3.2.1, provided the number of holders (barrels) are included for each set as specified in table II. The number of pens, and their sizes shall be as specified in table II. The set number and the point material requirements (see 3.3.1 - stainless steel, tungsten carbide, or synthetic jewel), shall be specified (see 6.2). A plastic bottle with a neck suitable for filling the pen cartridge with ink, and not less than one point section wrench shall be furnished with each set. Sets made up of pen size combinations other than the sets of table II may be specified, provided the sizes, the outside diameter of the point tip, and point material requirements are specified (see 6.2).

3.3.2.1 Alternate pen for type II (sets). Drawing point section pens shall be designed with a refillable ink reservoir (cartridge) which shall attach to the point housing section. The pens shall consist of a cap, a reservoir (cartridge), a point housing section, a point section (see 3.3.3), and an external and internal threaded locking ring which clamps the ink reservoir to the point housing section and engages a holder (barrel). The drawing point section pens shall be for use with drawing ink conforming to TT-I-[illegible]b and TT-I-[illegible], shall meet the features specified in 3.3.1.1 (except only one point section wrench is required per set), the performance requirements of 3.3.1.2, point material requirements of 3.3.1 and outside diameter at point tip as related to size designations of table I.

TABLE II

Set Number	Size							Number of Holder(s) 1/
	00	0	1	2	2½	3	3½ or 4	
1	x	x	x					1
2	x	x	x	x				1
3	x	x	x	x		x		2
4	x	x	x	x	x	x	x	3

1/ Applies to alternate pen described in 3.3.2.1.

3.3.2.2 Pen set case. A pen set case shall be provided with each pen set and shall provide storage for the pens and components. The case shall be designed with compartments or sleeves for the number of pens specified and the components of the set. Provisions shall be incorporated in the case to allow the pens to be placed in a point-up position during use.

3.3.3 Point section. The point section shall consist of an external threaded body with a built-in point tube, a safety plug, and a weighted cleansing wire. The threaded body shall be designed to join the point housing section. The appropriate size designation shall be marked on the section. When specified (see 6.2), replacement point sections shall be furnished as an assembled unit in the point material and sizes designated (see 3.3.1). A point section wrench shall be furnished with each replacement point section.

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3.4 Manufacturer's identification. The name of the manufacturer or his trademark shall be permently marked on each pen and pen set case.

3.5 Instructions. Directions for using the pen shall be furnished with each pen or each set.

3.6 Workmanship. All pens, including pen set case and pen set components, shall be free from defects which could impair their accuracy or servicability. The pens, pen set case and pen set components shall have no rough edges, burrs, cracks or chips.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. For purpose of sampling, a lot shall consist of all pens presented for inspection at one time as defined in MIL-STD-105.

4.2.2 Sampling of end-item for visual defects. A random sample of pens shall be taken from each lot in accordance with MIL-STD-105, inspection level I. The acceptance quality level (AQL) shall not exceed 4.0 percent defective. The sample unit shall be one technical fountain pen, or one set, as applicable.

4.2.3 Sampling for preparation for delivery requirements. For examination of interior packages the sample unit shall be one shipping container fully prepared for delivery, and selected at random just prior to the closing operation. Sampling shall be in accordance with MIL-STD-105. Closure defects shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end-item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

4.2.4 Sampling for end-item tests. A random sample of pens shall be taken from each lot in accordance with MIL-STD-105, inspection level S-2. The AQL shall not exceed 6.5 percent defective. The sample unit shall be one technical fountain pen, or one set, as applicable.

4.3 Examination.

4.3.1 Visual examination of end-item. Samples taken in accordance with 4.2.2 shall be examined for defects listed in table III.

TABLE III. Classification of defects

Examine	Defects
Design and Construction	Not style specified.
	Cap does not have a built-in liner.
	Color coding not as specified in 3.3.1.
	Point tube does not have an offset.
	Line size designation not stamped on pen barrel or cap.
	Point section wrench missing.
Pen set case (type II only)	Not as specified in 3.3.2.2.
Manufacturer's identification	Not marked on each pen and pen set case, as applicable.
Instructions	Missing (3.5).
Workmanship	Defects which could impair accuracy or serviceability.
	Rough edges, burrs, cracks or chips.

4.3.2 Examination of preparation for delivery. Samples taken in accordance with 4.2.3 shall be examined to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table IV.

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TABLE IV. Classification of preparation for delivery defects

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, or distortion of containers.

4.4 End-item tests. Samples taken in accordance with 4.2.4 shall be subjected to the following tests. Unless otherwise specified, tests shall be conducted in an atmosphere maintained at a temperature of 23° ±1.1° C. (73.5 ±2° F.) and a relative humidity of 50 ±4 percent. Test supplies and equipment, and the sample pens shall be conditioned in these atmospheric conditions for a minimum of 12 hours prior to tests. Tests involving writing shall be conducted with the pen held at an angle of 80 to 90° from the horizontal and with a writing pressure recommended by the manufacturer.

4.4.1 Performance. Drawing paper conforming to the requirements of UU-P-223, type I, class 1, style A, shall be used to test pens that have a stainless steel point. The matte side of drafting film conforming to the requirements of L-P-519, shall be used to test pens that have a tungsten carbide or synthetic jewel point.

The pen shall be filled with ink conforming to the requirements of TT-I-528 for use on drawing paper, and TT-I-533 for use on drafting film.

Using a straight edge, draw 10 lines, each 6 inches long (pretest lines may be drawn to actuate the ink flow). The pen shall be lifted vertically from the drawing material at the end of each line. The rate of speed to draw each line shall be 10 ±2 seconds. Tightly cap the pen after the last line is drawn and store in a point-up position.

After the inked lines have dried, they shall be examined for compliance with 3.3.1.2.

4.4.1.1 Startability. After the pens tested in 4.4.1 have been stored for 48 hours, repeat the test of 4.4.1, except the pens must start producing the first unbroken line within 15 seconds (includes priming time). The priming method (actuating ink flow) recommended by the manufacturer shall be followed.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Pens.

5.1.1.1 Unit packaging. Unless otherwise specified, each pen shall be packaged in accordance with the manufacturers standard practice, providing that this insures protection for the pens during shipment and provides safe delivery to destination.

5.1.1.2 Intermediate packaging. Unless otherwise specified, twelve unit packages shall be intermediate packaged in a close-fitting box conforming to PPP-B-566, PPP-B-636, PPP-B-665 or PPP-B-676. The box shall be closed and sealed in accordance with the applicable specification or appendix thereto.

5.1.2 Pen sets. Unless otherwise specified, each pen set shall be packaged in accordance with the manufacturers standard practice, providing that this insures protection for the pen sets during shipment and provides safe delivery to destination.

5.1.3 Replacement points.

5.1.3.1 Unit packaging. When specified, each point shall be packaged in accordance with the manufacturers standard practice, providing that this insures protection for the points during shipment and provides safe delivery to destination.

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5.1.3.2 Intermediate packaging. Unless otherwise specified, twelve replacement points shall be packaged in accordance with the manufacturers standard practice, providing that this insures protection for the points during shipment and provides safe delivery to destination.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Unless otherwise specified, 4 intermediate packages of pens, 12 packaged pen sets or 6 intermediate packages of replacement points shall be packed in a close-fitting box conforming to PPP-B-636, class weather-resistant. The box shall be closed, waterproof sealed and strapped in accordance with the appendix to PPP-B-636.

5.2.2 Level B. Unless otherwise specified, the level B packing shall be in accordance with 5.2.1 except that the boxes shall conform to PPP-B-636, class domestic.

5.2.3 Level C. Pens, pen sets, and replacement points packaged as specified in 5.1 shall be packed in containers that will assure carrier acceptance and safe arrival at destination in compliance with National Motor Freight Classification or the Uniform Freight Classification.

5.3 Marking. In addition to any special marking required by the contract or order, marking shall be in accordance with 5.3.1 or 5.3.2.

5.3.1 Civil Agencies. Marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The technical fountain pens covered by this specification are intended for use in freehand writing or drawing. Stainless steel points are for general use, and the tungsten carbide of synthetic jewel point are for coated drafting film (see 3.3.1).

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number and date of this specification.

b. Type and style of pens desired (see 1.2.1). When type I pens are desired, include style, size designation and point material requirements (see 3.3.1). When type II pens (sets) are desired, include set number and point material requirements (see 3.3.2).

c. Specify when replacement point assemblies only are desired. Include size designation and point material requirements (see 1.2.1 and 3.3.3).

d. Bid samples, if required, and quantity required (see 3.1).

e. Specify when pens other than size designation in table I are desired. Include style, size, the outside diameter of the point tip (including tolerance), and point material (see 3.3.1).

f. Specify when pen sets of styles or size combination other than those of 3.3.2 and table II are desired. Include size, the outside diameter of the point tip (including tolerance), components, and point material.

g. Level of packing required (see 5.2).

h. Any special marking required (see 5.3).

6.3 Pens above size 5 are designed without an offset on the point tube.

6.4 Bid samples. Bid samples submitted in accordance with 3.1 will be examined for:

Suitability of materials used.
General appearance and workmanship.

6

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, and mail to: Gen. Services Admin., FSS (FWSC), Washington, D.C. 20406

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

GG-F-001189 (GSA-FSS) - Fountain Pen, Technical

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

FORM GSA FEB 69 2200

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